Dear Editor,

On behalf of all authors, I am pleased to submit our manuscript entitled **"Transport-Coupled Bayesian Flows for Molecular Graph Generation"** for consideration as an Original Paper in *Bioinformatics*.

This work addresses molecular graph generation, a central problem in AI-driven molecular design and computational drug discovery. While many existing diffusion-based approaches operate by regressing continuous embeddings and then applying hard discretization during sampling, this creates a mismatch between training and generation for inherently discrete molecular graphs. Such a mismatch can limit structural fidelity, diversity, and sampling efficiency.

In this study, we propose **TopBF**, a transport-coupled Bayesian Flow framework for molecular graph generation. TopBF performs generation in continuous distribution-parameter space and uses a CDF-based decoding scheme to compute category probabilities directly, thereby aligning the training objective with the discrete decision process used at sampling time. In addition, we introduce a topology-aware Quasi-Wasserstein transport coupling under graph-geodesic costs, which encourages structure-consistent alignment of atom and bond categorical mass while preserving the closed-form Bayesian update mechanism. The framework also supports training-free, property-conditioned generation through guided sampling.

Experimentally, TopBF achieves strong performance on the standard QM9 and ZINC250k benchmarks, with improvements in structural fidelity and distributional quality relative to representative baselines, while requiring substantially fewer sampling steps. We further show that the proposed transport coupling contributes consistently to generation quality, and that the model supports effective controllable generation without retraining the base generative model.

We believe that this work is well aligned with the aims of *Bioinformatics*, as it presents a methodological advance in generative modeling with direct relevance to molecular representation learning, cheminformatics, and AI for drug discovery. More broadly, the study may be of interest to readers working on bioinformatics methods, molecular design, and machine learning for biomedicine.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree to its submission to *Bioinformatics*. The authors declare no conflict of interest.

Thank you for your time and consideration. We would be grateful for the opportunity to have this work considered for publication in *Bioinformatics*.

Best regards,
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